Classifying Vocal Biomarkers with Minsk2020 ALS Database

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Neurodegenerative disorders like ALS can be reflected in vocal biomarkers

- Amyotrophic lateral sclerosis (ALS) is a neurodegenerative disease for which vocal biomarkers show strong potential. As ALS progresses, it affects both upper and lower motor neurons, often leading to dysarthria, a motor speech disorder caused by impaired coordination of the bulbar muscles involved in speech production [3], [4].
- A prolonged vowel provides a quasi-periodic and consistent signal, and this consistency can reflect an individual's physiological muscle control and overall vocal health or disease. Physical health can be measured by quantifying characteristics of vocal signals rooted in physiology.

Neuro-muscular control can be measured by the shimmer, jitter, HNR, and per-second harmonic power contribution in a prolonged vowel

The objective was to design a system to quantify fundamental vocal features that are physiologically interpretable and can serve as informative indicators of vocal health. We utilized the open-source Minsk2020 ALS dataset, which contains 120 prolonged-vowel recordings approximately evenly split between ALS and non-ALS individuals to:

1. **Develop an algorithm that takes a sustained vowel recording as input and outputs three perturbation metrics (Jitter, Shimmer, Per-second Power Contribution) and one background metric (Harmonic to noise ratio (HNR)).**
2. **Evaluate effectiveness of metrics to accurately characterize ALS and non-ALS prolonged vowels in Minsk 2020 ALS Database [1].**

A prolonged vowel is a quasi-periodic signal

The two main components of voiced speech production are the glottal source and the vocal tract.

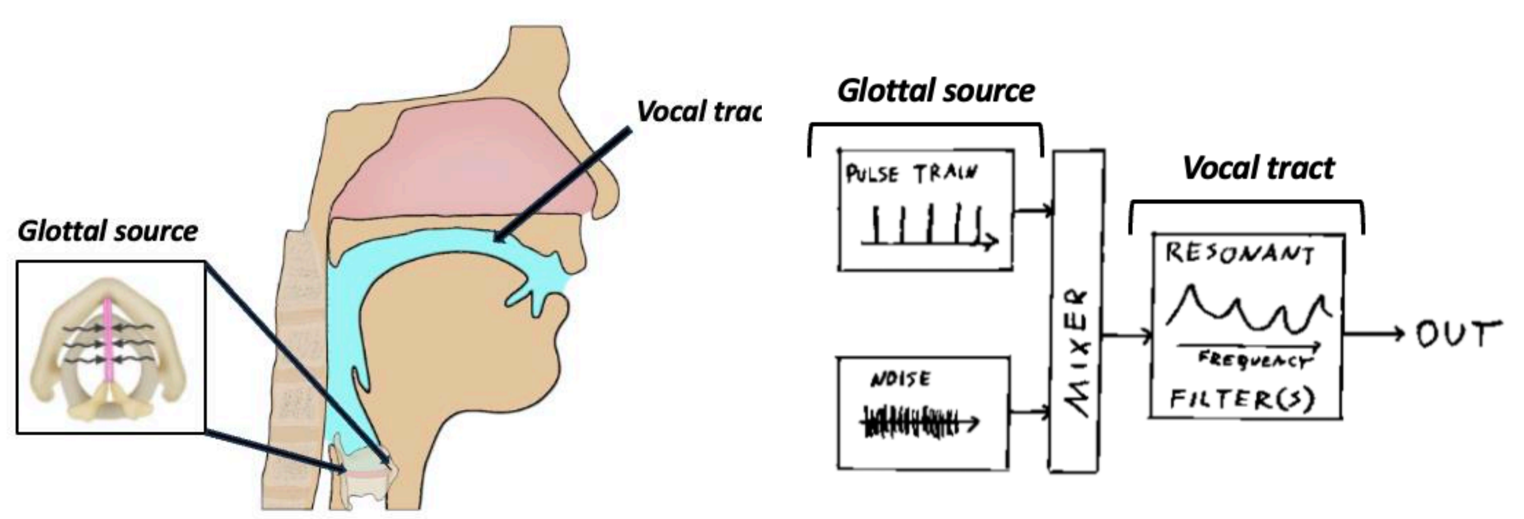
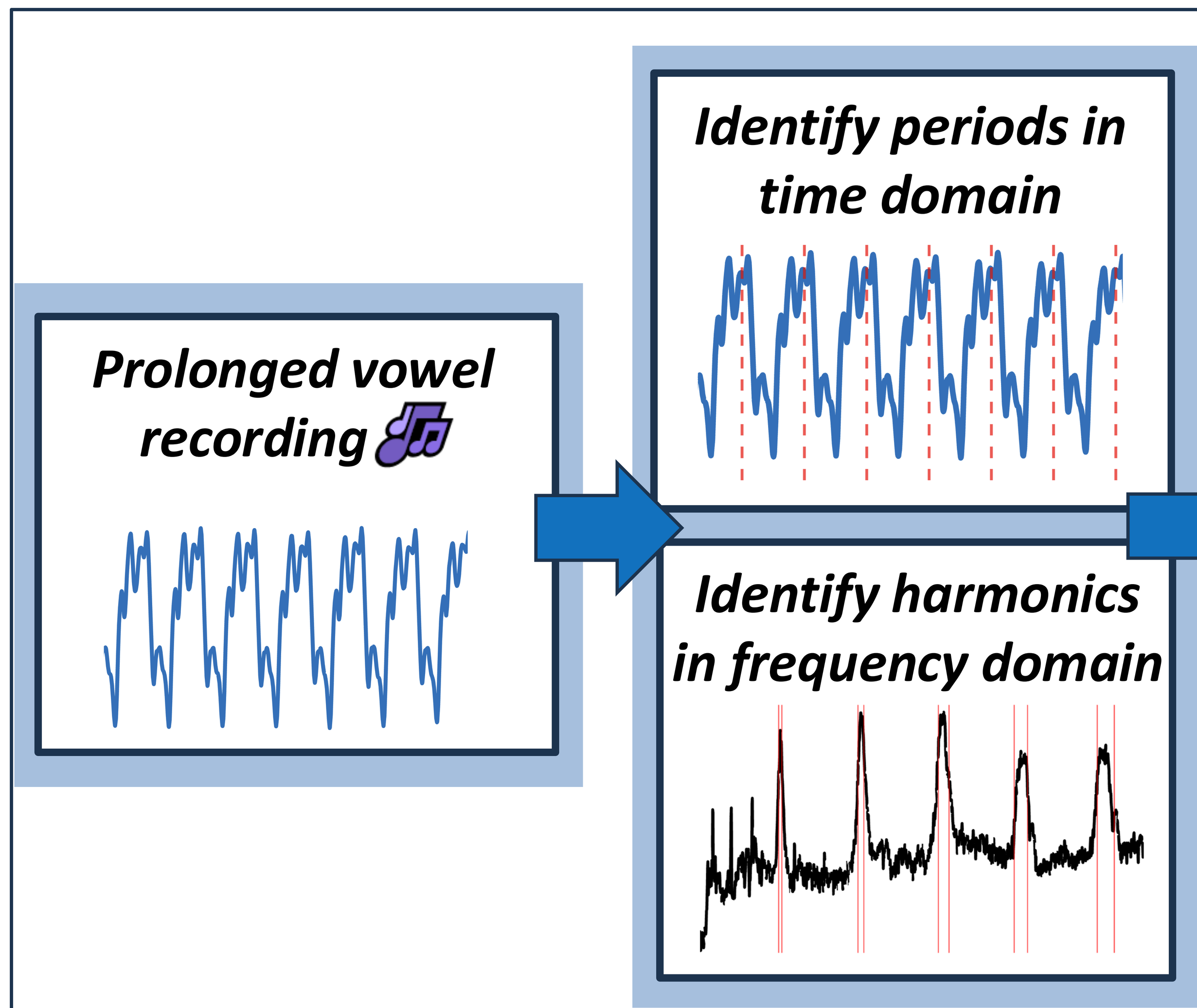


Figure 1: ***Signals and systems of phonation (right) [4], Anatomy of vocal system (left) [5]***

- The glottal source is generated by airflow passing through the vocal folds, which vibrate periodically at the fundamental frequency during phonation and can be approximated as a periodic impulse train.
  - Shimmer and jitter quantify pulse perturbation. Shimmer measures average amplitude variation across periods, while jitter measures period-to-period timing variation.
- The vocal tract acts as an acoustic resonator, shaping the sound by filtering the source signal and amplifying energy near its resonant frequencies, known as formants.
  - HNR measures the relative strength of harmonic energy compared to the noise floor, providing a measure of signal-to-noise in the vocal output.
- In theory, stable neuromuscular control during a prolonged vowel maintains a consistent vocal tract shape, so the resonant enhancement of harmonics remains stable over time.
  - Vocal tract perturbation can therefore be approximated by measuring how the power at the fundamental and harmonic frequencies changes over time.

[1] Mak-Sim, "GitHub - Mak-Sim/Minsk2020_ALS_database: Classification of ALS patients based on acoustic analysis of sustained vowel phonations," *GitHub*, 2025. https://github.com/Mak-Sim/Minsk2020_ALS_database (accessed Apr. 17, 2026).
[2] M. Vashkevich and Yu. Rushkevich, "Classification of ALS patients based on acoustic analysis of sustained vowel phonations," *Biomedical Signal Processing and Control*, vol. 65, p. 102350, Mar. 2021, doi: https://doi.org/10.1016/j.bspc.2020.102350.
[3] M. Larrazabal, "What als is and how it affects speech," Better Speech, https://www.betterspeech.com/post/what-als-is-and-how-it-affects-speech (accessed Feb. 6, 2026).
[4] "5. The Voice," *Ucsd.edu*, 2026. http://msp.ucsd.edu/syllabi/170.13f/course-notes/node5.html (accessed Apr. 15, 2026).
[5] "Harmonics Vs. Formants," *VoiceScienceWorks*, 2024. http://www.voicescienceworks.org/harmonics-vs-formants.html



Shimmer

$$= \frac{1}{N-1} \cdot \frac{\sum_{i=1}^{N-1} |A_{i+1} - A_i|}{\frac{1}{N}\sum_{i=1}^{N} A_i}$$

Jitter

$$= \frac{1}{N-1} \cdot \frac{\sum_{i=1}^{N-1} |T_{i+1} - T_i|}{\frac{1}{N}\sum_{i=1}^{N} T_i}$$

HNR

$$= 10 \log_{10} \frac{\left(\frac{H_p}{BW_h}\right)}{\left(\frac{N_p}{BW_n}\right)}$$

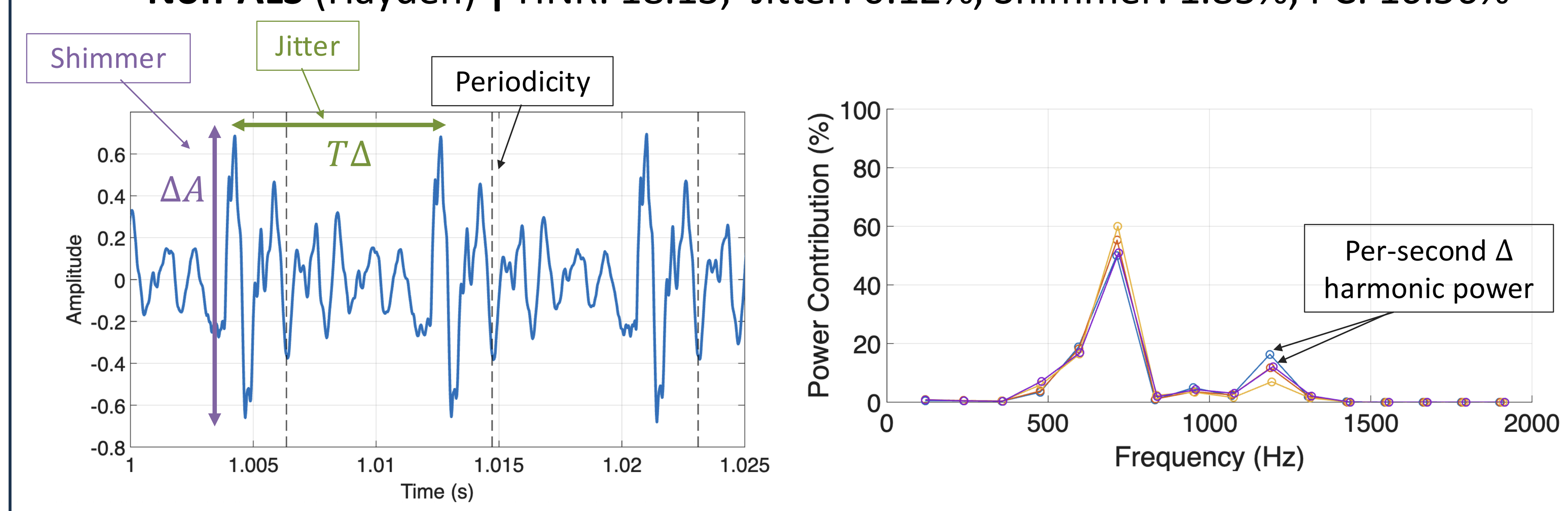
Per-second harmonic power

$$= \frac{\frac{1}{N_d}\sum d_s(k)}{\frac{1}{N_k}\sum PC_s(k)}$$

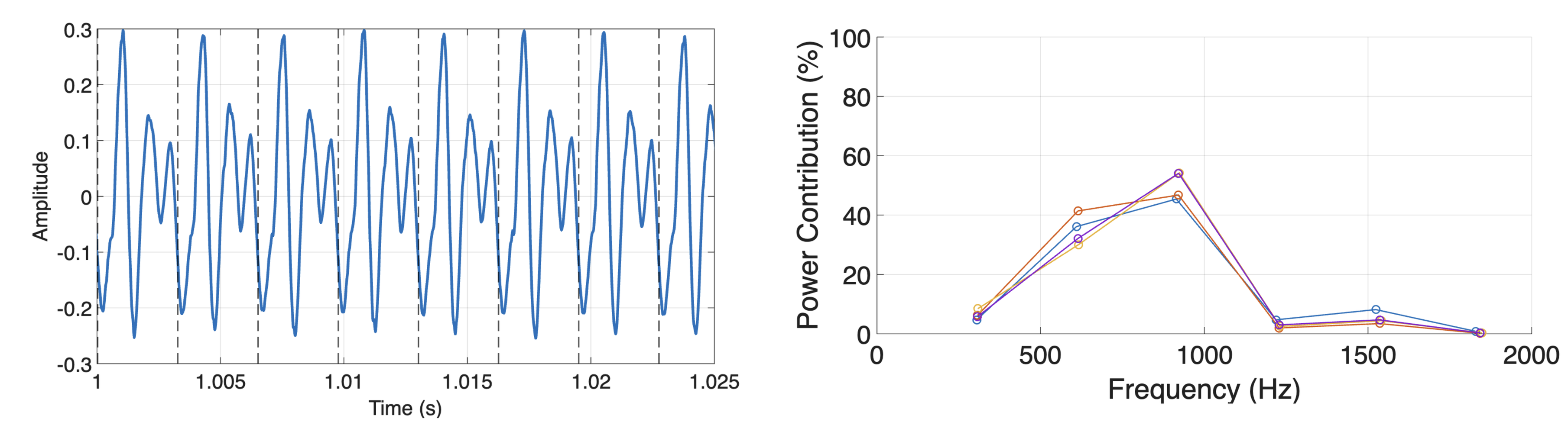
Our system inputs a spoken file and outputs biometrics!

Prolonged vowels differ per person and per condition.

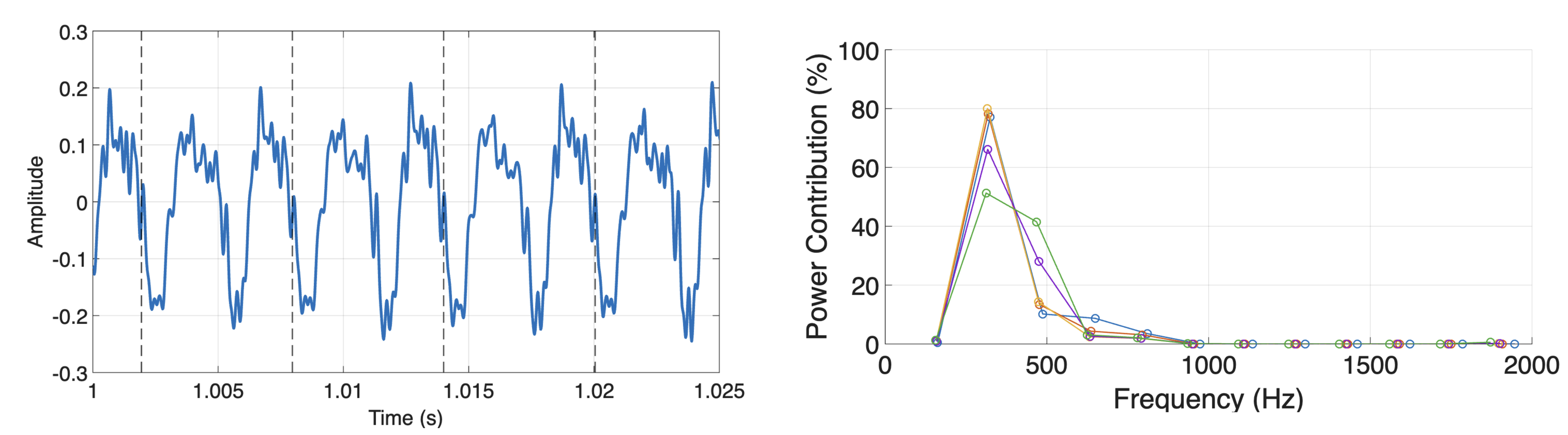
Non-ALS (Hayden) | HNR: 18.13, Jitter: 0.12%, Shimmer: 1.85%, PC: 16.56%



Non-ALS (Yvette) | 20.67 dB, Jitter: 0.05%, Shimmer: 1.74%, PC: 14.90%



ALS (Male, 67 years old) | HNR: 5.79 dB, Jitter: 0.35%, Shimmer: 2.23%; PC: 18%



ALS individuals exhibited higher vocal perturbation

ALS patients had higher values of all the perturbation metrics and lower values for the static metric. These results reflect physiological expectations; lower neuromuscular control results in more perturbation and lower HNR.

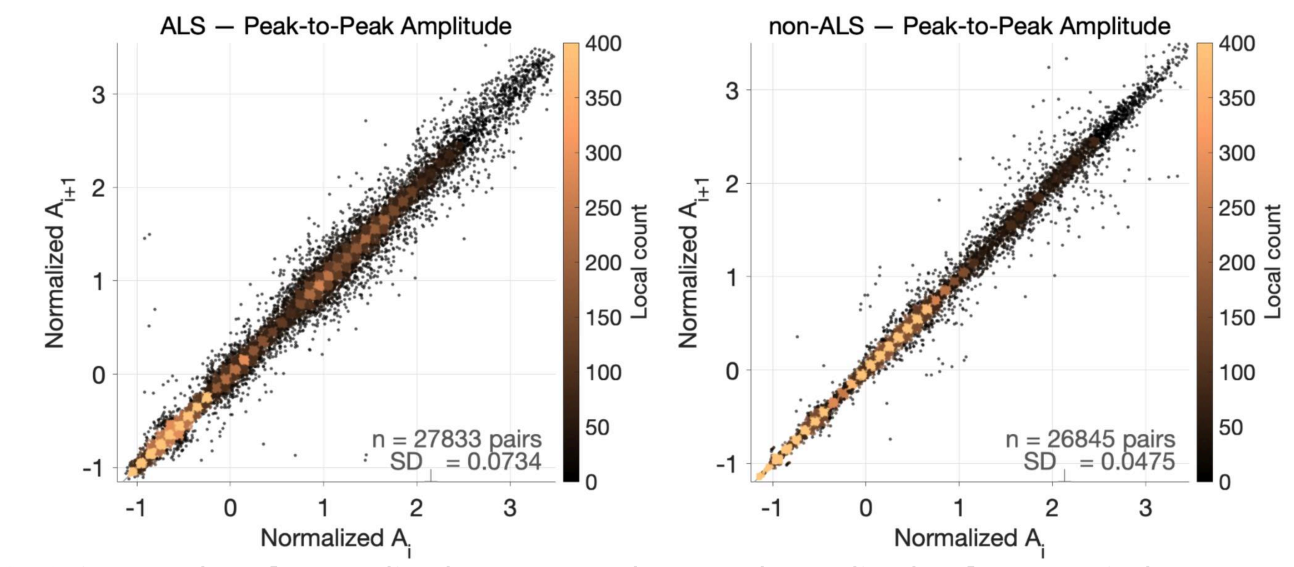


Figure 2: *Poincare plot of normalized z-score peak-to-peak amplitude of next period vs current period. ALS shown on right; Non-ALS on left.*

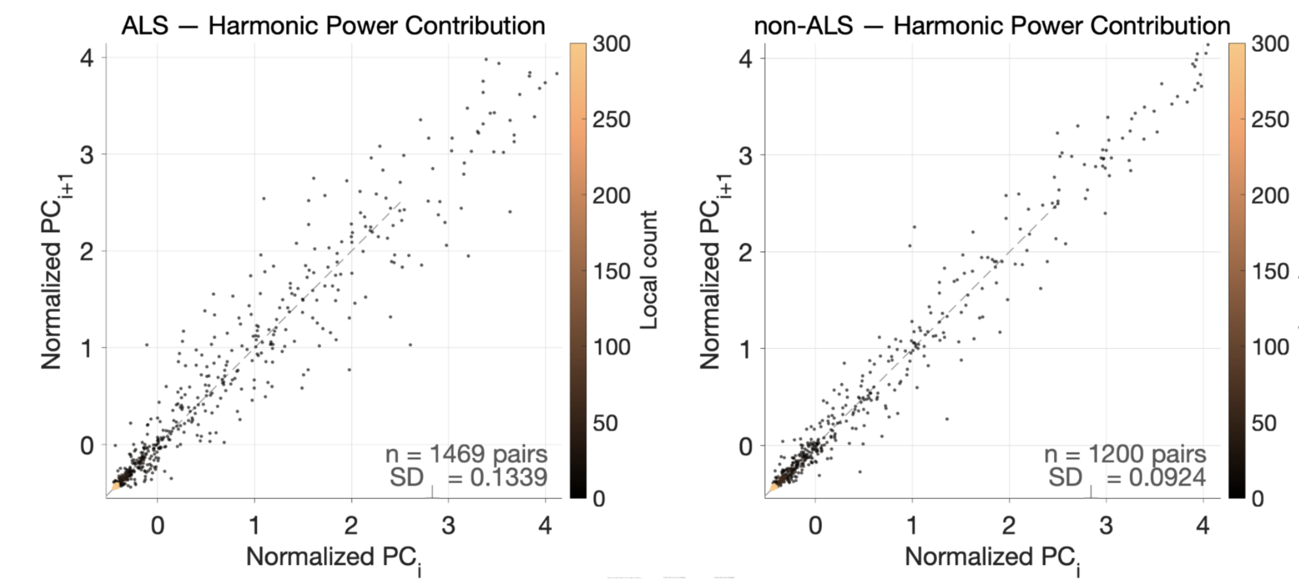


Figure 3: *Poincare plot of normalized z-score power contribution of next second vs current second. ALS shown on right; Non-ALS on left.*

Our vocal biomarkers characterized ALS and Non-ALS individuals in the Minsk dataset

Using a composite method of our metrics, our system characterizes ALS vs non-ALS individuals with AUC = 0.733.

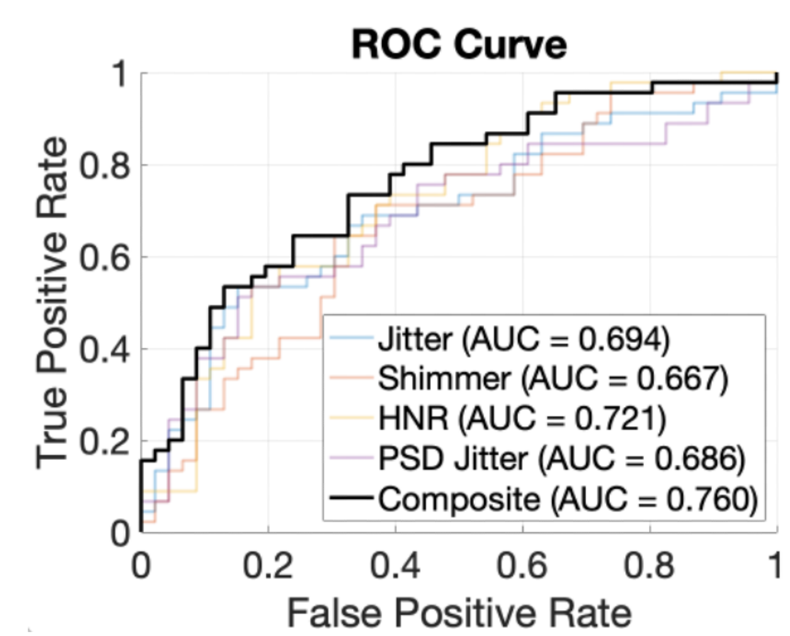


Figure 4: ROC describing characterization of ALS individuals from ALS and Non-ALS prolonged vowel files from the Minsk2020 ALS Database.

The assumption of a quasi-periodic signal is broken by highly degenerative vocals

- Two main limitations of our system arise from quantifying inconsistency relative to a periodic baseline. The method assumes the vocal signal is quasi-periodic and is therefore less robust for severely degraded voices.
- Algorithmically, these limitations appear in consecutive period detection and harmonic bandwidth estimation, which can in turn reduce the reliability of jitter, shimmer, HNR, and per-second harmonic contribution measurements.
- Overall, the program robustly computes vocal metrics including jitter, shimmer, HNR, and per-second power contribution. The code for these functions is publicly available and is intended to support future research on vocal biomarkers of disease and potential clinical applications.